

MATH 189 HW 1

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Question 1

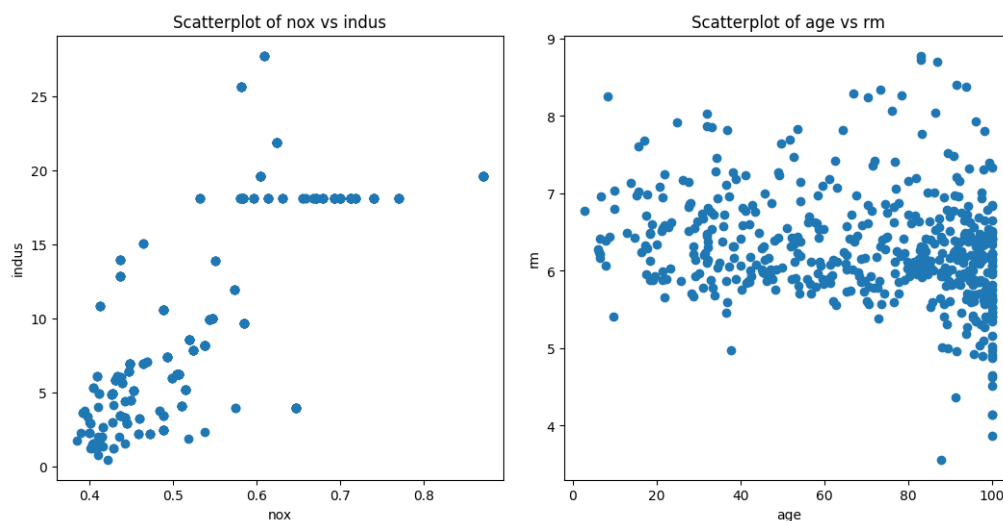
- a. Regression, $n = 500$, $p = 3$
- b. Categorical, $n = 20$, $p = 14$
- c. Regression, $n = 52$, $p = 3$

Question 2

- a. Obs. 1: 2
Obs. 2: 2
Obs. 3: $\sqrt{1^2 + 3^2} = \sqrt{10} \approx 3.16$
Obs. 4: $\sqrt{1^2 + 2^2} = \sqrt{5} \approx 2.23$
Obs. 5: $\sqrt{(-1)^2 + 1^2} = \sqrt{2} \approx 1.41$
Obs. 6: $\sqrt{1^2 + 1^2 + 1^2} = \sqrt{3} \approx 1.73$
- b. Our prediction for $K = 1$ is Green, because that's the color of the closest neighbor.
- c. Our prediction for $K = 3$ is Red, because it is close to two red neighbors and one green neighbors, so by plurality, red is the color of our prediction.
- d. If the Bayes decision boundary is highly nonlinear, then a low K value would provide the closest KNN decision boundary to Bayes. This is because low K values offer higher flexibility in the KNN decision boundary, which better fits a highly nonlinear Bayes decision boundary.

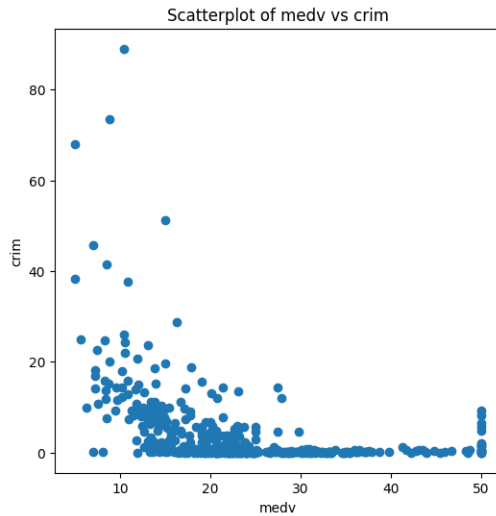
Question 3

- b. 506 rows and 13 columns. Rows represent each data point, columns represent the entries for each data point.



c.

For the first graph, I plotted nitrous oxides concentration versus proportion of non-retail business acres per town. There is a positive correlation between the two, although not a strong one, it also shows heteroscedasticity. For the second graph, I plotted average number of rooms per dwelling versus proportion of owner occupied unites built prior to 1940. There seems to be no correlation for these predictors.



- d. For the this graph, I plotted median value of owner-occupied homes versus per capita crime rate by town. There is a weak negative correlation between the two, although linear regression might not be appropriate here.
- e. The highest crime rate per capita is 88.9762 for town index 380, its range is 88.96988 which is a very wide range between the safest town versus the most dangerous town. The highest full-value property-tax rate is \$7,110,000 for town index 488, the range is \$5,240,000, which is again a very wide range between the areas with the highest property tax value and lowest property tax value. The highest pupil-teacher ratio is 22 for town index 354, and the range is 9.4, a moderate sized range.
- f. 35 suburbs in this data set bound the Charles river.
- g. The median pupil-teacher ratio among the towns in this data set is 19.05.
- h. Town index 398 has the lowest median value of owner occupied homes. The values of other predictors are 38.3518 for crime rate 0 for proportion of residential land zoned for lots over 25,000 sq.ft, 18.1 for proportion of non-retail business acres, does not bound the Charles river, 0.6930 for nitrogen oxides concentration, 5.453 for average number of rooms per dwelling. 100 for proportion of owner-occupied units built prior to 1940, 1.4896 for weighted mean of distances to five Boston employment centers, 24 for index of accessibility to radial highways, 666 for full-value property

tax rate per \$10,000, 20.2 for pupil-teacher ratio, 30.59 for lower status of population.

i.